

## EASTERN SIERRA, CA, USA

WMO: 724800

Lat: 37.371N

Lon: 118.358W

Elev: 4102

StdP: 12.64

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 23157

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 12            | 16.1       | 19.3 | -6.5                          | 4.5 | 43.8 | -2.7 | 5.5 | 46.2 | 28.4                  | 45.5 | 25.9 | 45.0 | 5.6                   | 320  | 0.561 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 39.3                   | 101.5           | 62.0 | 99.3 | 61.3 | 97.2 | 60.5 | 65.9                | 93.0 | 64.2 | 91.4 | 62.8 | 90.3 | 10.1                 | 160  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 57.3                            | 81.7 | 72.6 | 53.8 | 71.8 | 75.3 | 50.6 | 63.6 | 75.0 | 32.7          | 92.4 | 31.3 | 91.0 | 30.2 | 90.3 | 72.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |     |            |     |            |      |       |     |
|-------------------|------|------|----------------------------|-----|--------------------|-----|----------------------------------------------------|-----|------------|-----|------------|-----|------------|------|-------|-----|
|                   |      |      | Mean                       |     | Standard Deviation |     | n=5 years                                          |     | n=10 years |     | n=20 years |     | n=50 years |      |       |     |
| 1%                | 2.5% | 5%   | Min                        | Max | Min                | Max | Min                                                | Max | Min        | Max | Min        | Max | Min        | Max  |       |     |
| (a)               | (b)  | (c)  | (e)                        | (f) | (g)                | (h) | (i)                                                | (j) | (k)        | (l) | (m)        | (n) | (o)        | (p)  |       |     |
| (4) 25.5          | 22.5 | 19.5 | DB                         | 9.1 | 105.5              | 3.1 | 1.7                                                | 6.9 | 106.8      | 5.1 | 107.8      | 3.4 | 108.8      | 1.2  | 110.0 | (4) |
| (5)               |      |      | WB                         | 7.3 | 67.6               | 3.2 | 2.2                                                | 5.0 | 69.1       | 3.1 | 70.4       | 1.3 | 71.6       | -1.0 | 73.2  | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 57.1        | 39.8  | 42.3  | 49.2  | 54.6  | 62.9  | 72.4  | 78.1  | 75.7  | 68.1  | 56.9  | 45.5  | 38.2  | (6)  |  |
|                                                                        | DBStd       | 15.04       | 6.30  | 6.62  | 6.65  | 7.11  | 6.95  | 6.29  | 4.08  | 3.75  | 5.42  | 6.45  | 6.98  | 6.16  | (7)  |  |
|                                                                        | HDD50       | 1230        | 318   | 225   | 97    | 37    | 3     | 0     | 0     | 0     | 0     | 18    | 164   | 368   | (8)  |  |
|                                                                        | HDD65       | 4063        | 781   | 635   | 490   | 314   | 122   | 16    | 0     | 0     | 32    | 257   | 584   | 832   | (9)  |  |
|                                                                        | CDD50       | 3812        | 2     | 10    | 72    | 177   | 404   | 673   | 871   | 795   | 543   | 234   | 30    | 1     | (10) |  |
|                                                                        | CDD65       | 1170        | 0     | 0     | 0     | 4     | 58    | 239   | 406   | 331   | 125   | 7     | 0     | 0     | (11) |  |
|                                                                        | CDH74       | 23725       | 1     | 4     | 71    | 390   | 1685  | 4725  | 7130  | 6124  | 2984  | 589   | 22    | 0     | (12) |  |
|                                                                        | CDH80       | 13385       | 0     | 0     | 6     | 81    | 693   | 2745  | 4500  | 3730  | 1479  | 150   | 1     | 0     | (13) |  |
| (14) Wind                                                              | WSAvg       | 7.9         | 7.0   | 8.0   | 9.0   | 9.9   | 8.8   | 8.2   | 7.9   | 7.8   | 7.5   | 7.6   | 6.9   | 6.8   | (14) |  |
| (15) Precipitation                                                     | PrecAvg     | 5.4         | 0.9   | 1.0   | 0.4   | 0.2   | 0.2   | 0.2   | 0.2   | 0.2   | 0.2   | 0.1   | 0.6   | 0.7   | (15) |  |
|                                                                        | PrecMax     | 17.1        | 8.9   | 6.0   | 3.0   | 1.6   | 1.3   | 1.3   | 0.7   | 0.6   | 1.2   | 0.8   | 2.6   | 5.8   | (16) |  |
|                                                                        | PrecMin     | 1.8         | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | (17) |  |
|                                                                        | PrecStd     | 3.2         | 1.5   | 1.5   | 0.6   | 0.3   | 0.4   | 0.3   | 0.2   | 0.2   | 0.3   | 0.2   | 0.7   | 1.1   | (18) |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 69.2  | 73.6  | 80.3  | 86.5  | 95.1  | 102.9 | 105.1 | 102.5 | 97.0  | 88.6  | 76.9  | 67.3  | (19) |  |
|                                                                        |             | MCWB        | 46.2  | 47.2  | 50.8  | 54.1  | 58.5  | 61.1  | 63.5  | 63.1  | 60.6  | 56.2  | 50.3  | 45.6  | (20) |  |
|                                                                        | 2%          | DB          | 64.4  | 68.7  | 75.7  | 82.2  | 90.9  | 99.4  | 102.3 | 100.1 | 94.2  | 84.2  | 72.6  | 62.4  | (21) |  |
|                                                                        |             | MCWB        | 44.1  | 45.5  | 48.7  | 51.9  | 56.4  | 59.9  | 62.8  | 61.8  | 59.4  | 54.3  | 48.7  | 43.0  | (22) |  |
|                                                                        | 5%          | DB          | 60.4  | 64.6  | 72.2  | 78.8  | 87.5  | 96.8  | 100.1 | 98.0  | 91.9  | 80.5  | 68.4  | 58.4  | (23) |  |
|                                                                        |             | MCWB        | 42.6  | 44.1  | 47.4  | 50.7  | 55.1  | 59.4  | 61.9  | 60.8  | 58.5  | 52.5  | 46.7  | 41.4  | (24) |  |
|                                                                        | 10%         | DB          | 55.7  | 59.6  | 68.5  | 74.8  | 83.8  | 94.1  | 97.9  | 95.9  | 89.1  | 76.8  | 63.8  | 53.9  | (25) |  |
|                                                                        |             | MCWB        | 40.7  | 42.0  | 45.9  | 49.0  | 53.9  | 58.6  | 61.2  | 60.1  | 57.2  | 50.7  | 44.5  | 39.5  | (26) |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 49.5  | 49.2  | 52.2  | 56.1  | 60.4  | 65.5  | 68.4  | 67.8  | 64.3  | 59.2  | 52.2  | 48.1  | (27) |  |
|                                                                        |             | MCDB        | 64.8  | 67.7  | 76.2  | 82.5  | 90.8  | 97.1  | 96.2  | 96.1  | 89.6  | 77.0  | 73.1  | 62.4  | (28) |  |
|                                                                        | 2%          | WB          | 46.2  | 46.9  | 50.0  | 53.1  | 57.8  | 62.5  | 66.4  | 65.1  | 62.3  | 56.1  | 49.8  | 45.2  | (29) |  |
|                                                                        |             | MCDB        | 59.0  | 65.7  | 73.0  | 79.3  | 86.6  | 94.4  | 93.2  | 92.7  | 88.0  | 79.1  | 69.6  | 58.0  | (30) |  |
|                                                                        | 5%          | WB          | 44.1  | 44.9  | 48.3  | 51.2  | 56.3  | 60.9  | 64.9  | 63.5  | 60.4  | 53.8  | 47.7  | 42.4  | (31) |  |
|                                                                        |             | MCDB        | 57.3  | 61.8  | 70.2  | 76.5  | 84.1  | 92.8  | 91.4  | 91.3  | 86.1  | 77.4  | 66.2  | 55.6  | (32) |  |
|                                                                        | 10%         | WB          | 41.8  | 42.9  | 46.4  | 49.7  | 54.6  | 59.4  | 63.5  | 62.0  | 58.4  | 51.7  | 45.5  | 40.2  | (33) |  |
|                                                                        |             | MCDB        | 55.1  | 57.8  | 67.0  | 73.6  | 81.1  | 90.6  | 90.1  | 89.6  | 84.9  | 74.2  | 62.9  | 53.0  | (34) |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 29.4  | 29.6  | 33.0  | 33.3  | 35.0  | 38.6  | 39.3  | 40.5  | 39.3  | 36.6  | 33.8  | 29.5  | (35) |  |
|                                                                        |             | MCDBR       | 37.2  | 38.6  | 39.1  | 40.0  | 40.6  | 42.2  | 41.9  | 42.6  | 42.6  | 41.9  | 39.6  | 36.9  | (36) |  |
|                                                                        | 5% WB       | MCWBR       | 20.7  | 19.7  | 18.6  | 17.3  | 15.6  | 14.9  | 14.7  | 15.6  | 17.3  | 19.2  | 20.7  | 21.3  | (37) |  |
|                                                                        |             | MCWBR       | 32.6  | 34.6  | 37.1  | 38.0  | 38.7  | 39.9  | 36.7  | 38.2  | 39.6  | 38.2  | 37.3  | 33.6  | (38) |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.262 | 0.276 | 0.287 | 0.318 | 0.333 | 0.330 | 0.340 | 0.328 | 0.305 | 0.289 | 0.265 | 0.261 | (40) |  |
|                                                                        |             | taud        | 2.473 | 2.446 | 2.443 | 2.398 | 2.415 | 2.437 | 2.470 | 2.478 | 2.545 | 2.567 | 2.576 | 2.513 | (41) |  |
|                                                                        | Ebn at Noon |             | 298   | 306   | 312   | 305   | 300   | 300   | 296   | 298   | 302   | 298   | 297   | 292   | (42) |  |
|                                                                        |             | Edh at Noon | 25    | 30    | 33    | 37    | 37    | 36    | 35    | 34    | 30    | 26    | 23    | 23    | (43) |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 857   | 1146  | 1571  | 2014  | 2292  | 2636  | 2454  | 2307  | 1962  | 1446  | 1013  | 764   | (44) |  |
|                                                                        | RadStd      |             | 74    | 107   | 107   | 107   | 177   | 168   | 148   | 125   | 75    | 97    | 56    | 76    | (45) |  |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |      |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |      |
|      |                         | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |      |
| (46) | Station Only            | +0.68 | N/A     | -2.63  | N/A     | N/A   | N/A   | N/A         | N/A   | +220  | +146  | (46) |
| (47) | Regional (14 neighbors) | +0.97 | N/A     | N/A    | +0.72   | N/A   | N/A   | -81         | N/A   | +274  | +166  | (47) |

Nomenclature: See separate page